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WeLearn in Singapore and Paris – a UN SDG 4 Project

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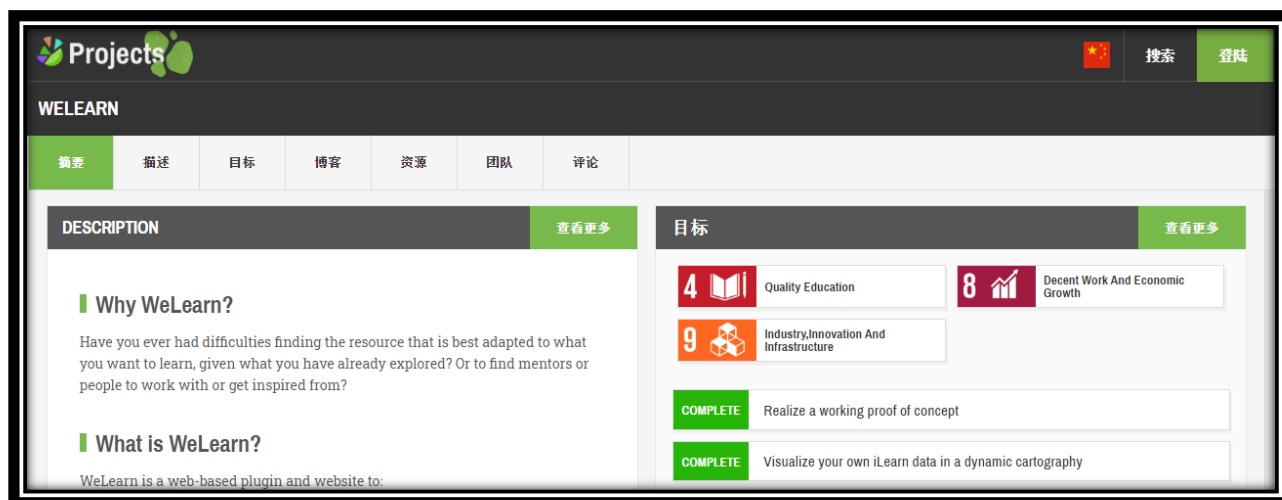
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Introduction

As education has become more prevalent in the 21st century, the focus of learning has moved into sustainability- to ensure equal opportunities for lifelong education. Within this digital age when information is readily available with a click of the mouse, the paradigm of learning has shifted drastically. Hence, in conjunction with the WeLearn team in CRI Paris at University of Paris, the National University of Singapore (NUS) "Learning to Learn Better Team" aims to tackle the problem of sustainable education for all through the collaborative platform of WeLearn.



Our Course

Among the myriad of skills and knowledge in university, the [Learning to Learn Better](#) module (ALS1010) teaches students learning strategies derived from the fundamental sciences. With psychology as its core, [the course](#) aims to teach students effective rules to accelerate their learning potential.

“In today’s fast-changing world, it’s not so much what you know anymore that counts, because often what you know is old. It is how fast you learn. That skill is priceless.”

— Robert T. Kiyosaki (Best-selling Author of *Rich Dad Poor Dad*)

We believe that teaching students to adopt lifelong learning strategies would enable them to stay relevant and competitive and prepare them for the future. By underlying the fundamental process of learning, we hope to provide opportunities for growth and sustainable lifelong learning. Hence, in collaboration with the [WeLearn](#) team, the ALS1010 team are thrilled to hit new learning potentials through the WeLearn application.



Instructors for ALS1010 preparing for the sessions. Photo courtesy of TAN Hui Ru and Magdelaine NG.

WeLearn project

Despite the rapid improvements in technology, an estimated 260 million children all around the world are out of school. <http://data.uis.unesco.org/>. Furthermore, many university students are burdened with student loan once they graduate. We believe that a community of learners can help provide support and necessary resources, bringing us closer towards the goal of sustainable quality education. In partnership with the French development agency and UNESCO, WeLearn aims to build a community of learners. Potentially the ‘Facebook’ for learning, users can share ideas with one another and network with possible academic experts. Users of similar interests can discuss shared interests and find mentors and mentees in their respective fields.

As part of the global digital learning ecosystem, WeLearn functions as a network-based plugin to organize learning resources by concepts and share the information with other users. In addition, the web-based extension provides categorization of articles based on complexity ranging from easy to difficult. This function allows the user to determine their level of competency in their topic of choice. We made a video guide to illustrate how learners can join this SDG project as a global learner. This video guide can be found [here](https://youtu.be/DjAiOoSvu2k) and is open to the public.

URL: <https://youtu.be/DjAiOoSvu2k>



Sustainable Development Goal 4

Photo courtesy of Sustainable Development UN

Why it matters?

As part of the project, our students identify three different articles and save it in the WeLearn page using the extension. Following which, they referenced two articles from their peers in the WeLearn NUS group. The students were tasked with an assignment to write out their thoughts of reflections regarding the articles they have chosen and the takeaways they had from using the application.

Popular Topics of interest

Among the 1000 different articles highlighted by the students, there were a few concepts which stood out, specifically the coronavirus, the economy and feminism. From the experiment, there was a diverse selection of niche topics from brown bears to the economics of the FIFA world. A substantial number of students bookmarked articles related to their course of study, some of which were academic. However, it was fascinating how students from different majors were able to reference similar knowledge from other fields and apply it to their course of study. It was heartening to see students referring to articles taught in the ALS module while providing thought-provoking insights in their reflections.



A screenshot of the WeLearn interface

Examples of learners' work

Toh You Jing

Why the best material for a homemade coronavirus face mask is hard to identify

- Experts think people pass the virus to others primarily through respiratory droplets.
- Tightly fitting N95 masks are designed to protect the wearer by trapping infectious particles in complex layers of extremely fine polypropylene fibers.
- These fibers are also electrostatically charged to provide extra "stickiness" while retaining breathability.
- Natural fibers can swell when exposed to moisture from a person's breath, changing the fabric's performance in unpredictable ways.
- Hence, homemade masks using natural fibers must compensate by using multiple layers of fabric.
- This brings about the next problem which is: the inherent trade-off between the size of pores in the fabric and breathability—the least porous materials will also be harder to breathe through.
 - I can relate to this point. Especially with my older relatives, they find themselves short of breath just minutes after wearing the cloth mask as compared to a surgical mask.



Similar topic: Coronavirus pandemic: Tracking the global outbreak

- Coronavirus is continuing its spread across the world, with more than 6.7 million confirmed cases in 188 countries. Nearly 400,000 people have lost their lives.
- WHO has warned that people should be prepared for new outbreaks to build up very quickly - especially in areas where lockdowns are eased.
 - This would be relevant to Singapore as we have just started Phase I and planning to transit to Phase II and III. We are seeing an increase in local cases to double digits on the 6th day of Phase I. However, it remains to be seen the effects of easing the measures due to the 14 day incubation period.
- Globally, at least 4.5 billion people - half the world's population - have been living under social distancing measures, according to the AFP news agency's estimates.
- With more than 1.8 million cases, the US has the highest number of confirmed infections in the world. It has also recorded over 109,000 deaths.

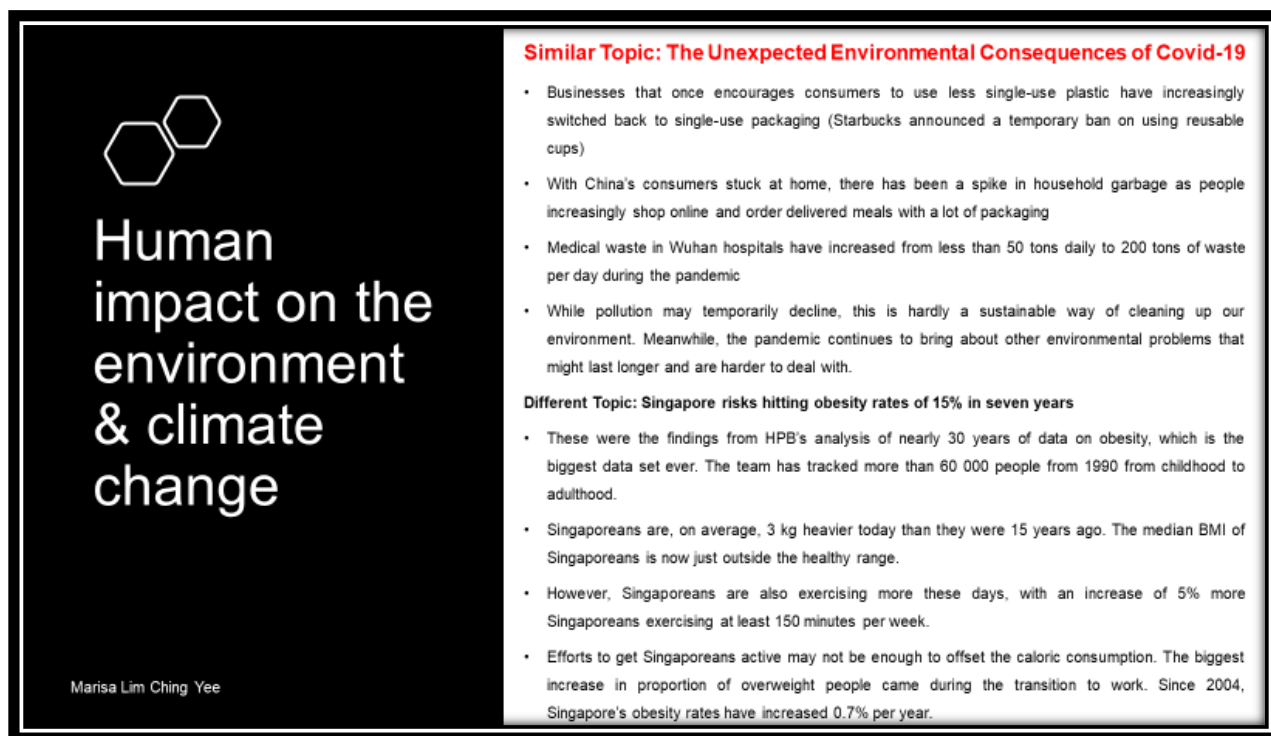
Different topic: How video games affect the brain

- Results of 116 scientific studies indicate that playing video games not only changes how our brains perform but also their structure.
- The benefits of playing video games include:
 - Improvements in several types of attention, including sustained attention and selective attention.
 - Increases the size and competence of parts of the brain responsible for visuospatial skills - a person's ability to identify visual and spatial relationships among objects.
- However, video games can be addictive. In gaming addicts, there are functional and structural alterations in the neural reward system. Changes that are also seen in other addictive disorders.
- For older adults, video games may improve significant cognitive areas such as working memory and sustained attention.
- "If the target is to improve older adults' cognitive control, reasoning, and higher-order cognitive skills, and stave off dementia and Alzheimer's disease as long as possible, then maybe strategy games are the way to go."
 - In my personal experience of volunteering at AWWA Senior Activity Centre, they have collaborated with Temasek Polytechnic to create a digital supermarket game for the elderly to play and hone their cognitive abilities.




Content summary of WeLearn project

Photo courtesy of Toh You Jing.



Human impact on the environment & climate change

Marisa Lim Ching Yee

Similar Topic: The Unexpected Environmental Consequences of Covid-19

- Businesses that once encourages consumers to use less single-use plastic have increasingly switched back to single-use packaging (Starbucks announced a temporary ban on using reusable cups)
- With China's consumers stuck at home, there has been a spike in household garbage as people increasingly shop online and order delivered meals with a lot of packaging
- Medical waste in Wuhan hospitals have increased from less than 50 tons daily to 200 tons of waste per day during the pandemic
- While pollution may temporarily decline, this is hardly a sustainable way of cleaning up our environment. Meanwhile, the pandemic continues to bring about other environmental problems that might last longer and are harder to deal with.

Different Topic: Singapore risks hitting obesity rates of 15% in seven years

- These were the findings from HPB's analysis of nearly 30 years of data on obesity, which is the biggest data set ever. The team has tracked more than 60 000 people from 1990 from childhood to adulthood.
- Singaporeans are, on average, 3 kg heavier today than they were 15 years ago. The median BMI of Singaporeans is now just outside the healthy range.
- However, Singaporeans are also exercising more these days, with an increase of 5% more Singaporeans exercising at least 150 minutes per week.
- Efforts to get Singaporeans active may not be enough to offset the caloric consumption. The biggest increase in proportion of overweight people came during the transition to work. Since 2004, Singapore's obesity rates have increased 0.7% per year.

Content summary of WeLearn project

Photo courtesy of Marisa Lim Ching Yee

Like Marisa and You Jing, many of our student's research emphasised on good health and well-being, and climate action.

Acknowledgements

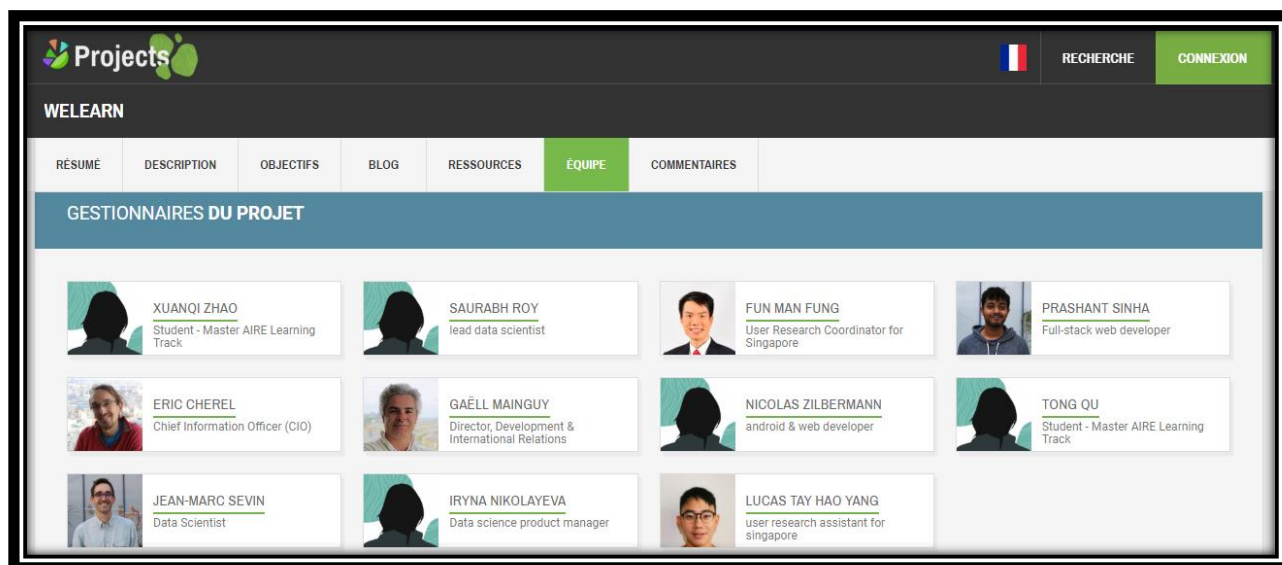
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Link: <https://projects.cri-paris.org/projects/Jq2tzEMD/team>

<https://projects.cri-paris.org/projects/Jq2tzEMD/summary>

International Younger Chemists Network (IYCN)

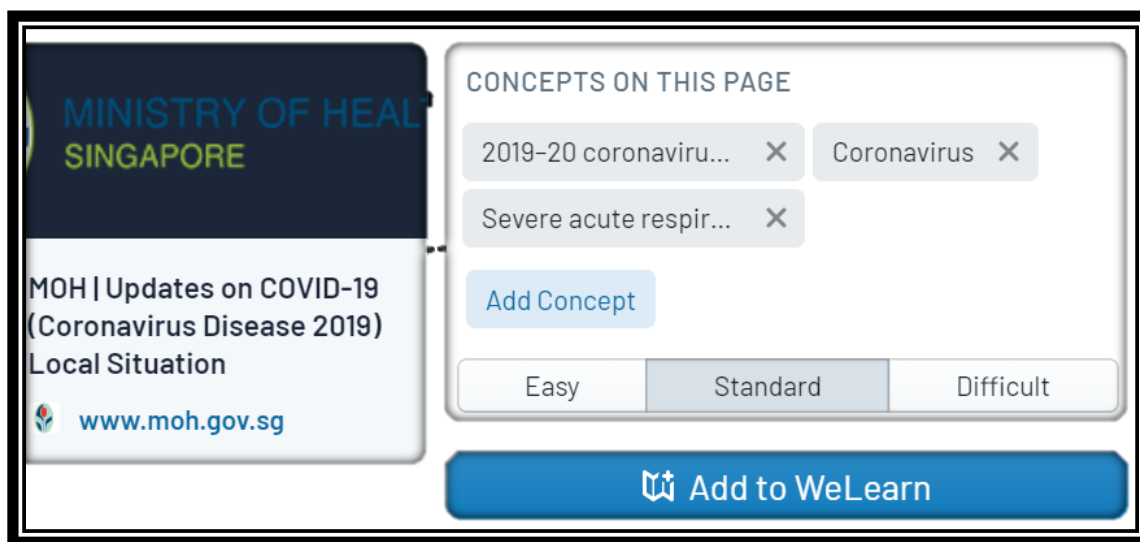
"Connect and empower younger chemists globally"



#WeLearntogether

1. Install the WeLearn extension
2. Bookmark your favourite articles on WeLearn (see [video guide](#))

<https://youtu.be/DjAiOoSwu2k>



3. Add relevant concepts and choose the level of difficulty for concepts
4. Find similar articles or different articles contributed by the community



References:

<https://welearn.cri-paris.org/pages/discover.htm>